

INDIA PORT LEVEL EXPORT ANALYTICS

(2018–2024)

An Exploratory Data Analysis of India's Export Performance Across States, Ports, Countries, and Principal Commodities Using Python

A PROJECT REPORT

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Course Name

AI Driven Data Analyst Certification – Entri Elevate

Dataset Source

India Data Portal / Directorate General of Commercial Intelligence and Statistics (DGCI&S)

Dataset Name

Exports at Principal Commodity Level

Study Period

2018–2024

September 2026

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1. Introduction

- **International trade data analytics** is an important application of data analysis for understanding export performance, identifying major contributors, and supporting data-driven business decision-making. India's export sector involves multiple states, ports, destination countries, and principal commodities, resulting in a large and complex volume of trade data. Analysing these dimensions together can provide a clearer understanding of the country's export structure and changing trade patterns.
- This project, **India Port Level Export Analytics (2018–2024)**, focuses on analysing India's export performance using the **Exports at Principal Commodity Level** dataset. The dataset, sourced from the **India Data Portal / Directorate General of Commercial Intelligence and Statistics (DGCI&S)**, contains detailed export records covering the period **2018–2024**. The data includes information on export date, state, port, destination country, principal commodity, quantity, and export value in both USD and INR.
- The project was undertaken as a **Hands-on Data Analytics Capstone Exercise** to apply the end-to-end data analytics process to a large-scale real-world dataset. The workflow includes data understanding, data cleaning and pre-processing, missing-value treatment, duplicate validation, data type conversion, feature engineering, exploratory data analysis, statistical summarization, and visualization.
- The analysis examines India's export performance from multiple perspectives, including **overall export value and quantity, major principal commodities, state-wise performance, yearly trends, commodity–state relationships, and changes in commodity performance across years**. It also evaluates the contribution and concentration of major commodities and states in overall exports.
- The project uses **Python and its data analytics ecosystem**, including Pandas, NumPy, Matplotlib, Seaborn, and Plotly, to transform the raw export records into structured analytical results and meaningful visual insights. The findings are intended to provide a clear understanding of the major patterns and variations present in India's exports during the study period and to support data-driven interpretation of export performance.

2. Problem Statement

India's export dataset contains millions of records covering multiple years, states, ports, destination countries, and principal commodities. With export values and quantities varying significantly across these dimensions, it is difficult to identify the major contributors, understand changes in performance, and assess the overall structure of India's exports by reviewing the raw data alone.

A structured analysis is therefore required to evaluate India's export performance during **2018–2024**, identify the leading commodities and exporting states, track yearly changes, and understand relationships between commodities and states. This project addresses this need by applying **data cleaning, exploratory analysis, statistical analysis, and visualization** to convert the detailed export data into clear and actionable business insights.

3. Project Objectives

The project aims to achieve the following objectives:

1. **Assess Overall Export Performance**
Analyze India's overall export performance across the available years using export value and quantity.
2. **Identify Top-Performing Commodities**
Identify the principal commodities that contribute significantly to India's total export value.
3. **Evaluate State-Wise Export Performance**
Analyze export performance across states and identify the major exporting states.

4. **Examine Yearly Export Trends**
Track changes in export performance across years and identify periods of growth, decline, and significant variation.
5. **Analyze Commodity–State Relationships**
Examine the relationship between principal commodities and states to identify commodities that contribute significantly to exports from different states.
6. **Identify Changes in Commodity Performance**
Analyze changes in commodity-wise export performance across different years.
7. **Measure Export Contribution and Concentration**
Measure the contribution and concentration of major commodities and states within India's overall exports.

4. Dataset Description

The project uses the **Exports at Principal Commodity Level** dataset to analyse India's export performance across different geographical, commodity, and time dimensions. The dataset contains detailed export records covering states, ports, destination countries, principal commodities, quantities, and export values.

4.1 Dataset Overview

Attribute	Details
Dataset Name	Exports at Principal Commodity Level
Source	India Data Portal / Directorate General of Commercial Intelligence and Statistics (DGCI&S)
Domain	Commerce / International Trade / Exports
Geographical Coverage	India
Study Period	2018–2024
Frequency	Monthly
Granularity	State-level export records
Total Records	6,912,492
Number of Original Columns	12
Key Dimensions	Date, State, Port, Destination Country, Principal Commodity
Key Measures	Quantity, Export Value (USD), Export Value (INR)
Additional Features Created	Year, Month, Month Name

4.2 Dataset Attributes Description

Attribute	Description
id	Unique identifier for each export record.
date	Date associated with the export record.
state_name	Name of the Indian state associated with the export activity.
state_code	Code representing the respective state.
country	Destination country for the exported goods.
port	Port through which the export was recorded.
pc_code	Code assigned to the principal commodity.
commodity	Name of the principal commodity exported.
units	Unit of measurement used for the exported quantity.
quantity	Quantity of the commodity exported.
dollars_value	Export value expressed in US Dollars (USD).
inr_value	Export value expressed in Indian Rupees (INR).
year	Year extracted from the date column for yearly analysis.
month	Month number extracted from the date column.
month_name	Name of the month extracted from the date column.

5. Tools and Technologies Used

The following tools and technologies were used throughout the project for data processing, analysis, visualization, and project documentation.

Tool / Technology	Purpose in the Project
Python	Core programming language used for data analysis and processing.
Google Colab	Cloud-based environment used to develop and execute the Python analysis workflow.
Pandas	Used for data loading, cleaning, transformation, aggregation, and analysis.
NumPy	Used for numerical operations and statistical calculations.
Matplotlib	Used to create static data visualizations.
Seaborn	Used to create statistical and exploratory data visualizations.
Plotly	Used to create interactive visualizations and relationship analysis.
GitHub	Used for project version control and portfolio sharing.
Power BI	Used for dashboard development and business intelligence visualization.

6. Data Cleaning and Preprocessing

6.1 Missing Value Treatment

- Missing **values were identified** across all dataset columns using null-value checks to determine the extent and location of missing data.
- The **unit's column contained 4,468,611 missing records (64.65%)**. Further analysis showed that these records had **zero quantity**, indicating that the missing unit values were associated with records where no measurable quantity was reported.
- To maintain data completeness, the missing values in units were **imputed with the mode, Kgs**, while the original quantity, export value, and other dataset values were retained without modification.

6.2 Duplicate Handling

- Duplicate records were checked across the dataset, and no duplicate records were identified, ensuring that each export record was retained as a unique observation.

6.3 Data Type Conversion

- The id **column** was converted to **string** to ensure consistent handling of record identifiers.
- The date **column** was converted to **datetime format**, enabling accurate time-based analysis and extraction of year and month.
- The state_code **column** was converted to **string** to preserve state codes consistently and prevent unintended numeric interpretation.

6.4 Feature Engineering

The **date** column was used to create time-based features to support yearly and monthly export analysis.

New Feature	Derived From	Purpose
Year	date	Year-wise export trend analysis
Month	date	Monthly export analysis
Month Name	date	Easy interpretation of monthly patterns

7. Exploratory Data Analysis and Statistical Summary

A final exploratory inspection was performed on the cleaned dataset to understand its overall structure, numerical characteristics, and data distribution. The analysis included descriptive statistics, measures of central tendency and dispersion, and distribution analysis of the key numerical variables: **Quantity, Export Value (USD), and Export Value (INR)**.

7.1 Descriptive Statistical Summary

The following table summarizes the key statistical measures for the three primary numerical variables: **Quantity, Dollars_Value, and INR_Value**.

Measure	Quantity	Dollars_Value (USD)	INR_Value (₹)
Mean	64,414.45	\$357,966.10	₹27,242,250
Median	0.00	\$19,502.00	₹1,466,100
Mode	0	\$1.00	₹82
Minimum	0	\$0.00	₹1
Maximum	7,620,654,000	\$1,449,331,220.00	₹119,263,437,035
Range	7,620,654,000	\$1,449,331,220.00	₹119,263,437,034
Variance	1.38×10^{13}	2.82×10^{13}	1.66×10^{17}
Standard Deviation	3,720,349	\$5,306,167	₹407,425,200

Key Observation: The large difference between the **mean and median**, together with the high **range and standard deviation**, indicates substantial variation and a **highly right-skewed distribution** across the export records **with extreme high-value observations**.

7.2 Distribution Analysis

The distribution of **Quantity**, **Dollars_Value**, and **INR_Value** was examined to understand the spread and concentration of export records. Due to the presence of highly skewed values and extreme observations, a **log1p transformation** was applied to improve the visual representation of the distributions. The transformation was used **only for visualization**, while the original dataset values remained unchanged.

Log-Transformed Distribution of Export Variables

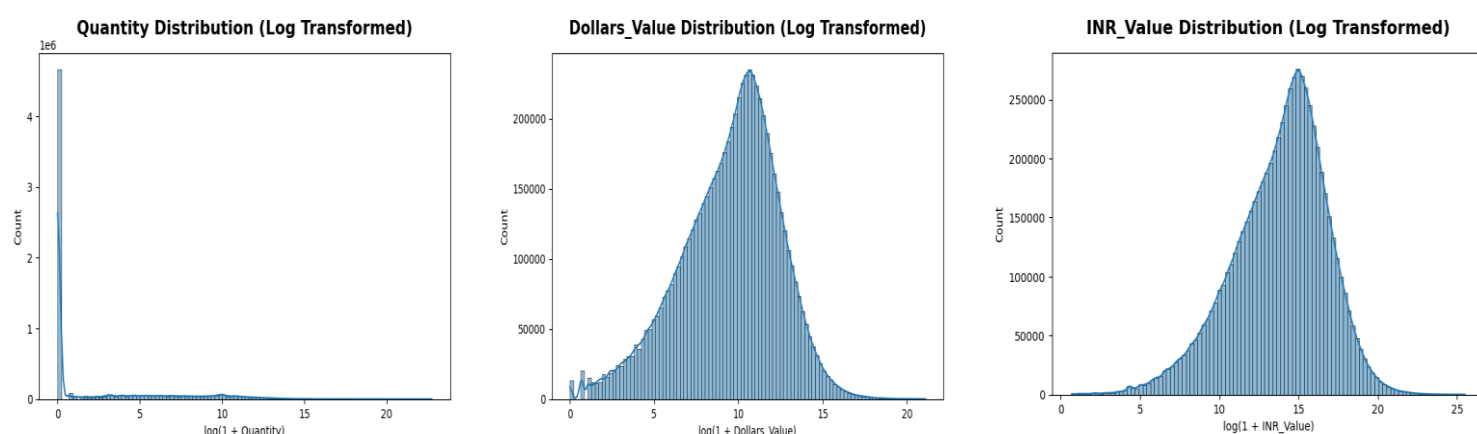


Figure 7.1: Log-Transformed Distribution of Key Export Variables

Interpretation

- **Quantity:** The distribution remains concentrated near zero even after transformation, reflecting the large number of records with **zero or very small quantities**.
- **Dollars_Value:** The log transformation substantially reduces the effect of extreme values and makes the distribution **more balanced and easier to interpret**.
- **INR_Value:** Similar to Dollars_Value, the transformation reduces the influence of extreme export values and provides a **more balanced distribution**.

8. Data Visualization and Objective-Wise Analysis

Each objective is presented with its **analytical approach**, **supporting visualization**, **interpretation**, **key finding**, and **business insight**. The analysis is based on the cleaned export dataset and the visualizations generated during the project.

8.1 Objective 1: Overall Export Performance

Assess Overall Export Performance

Analyze India's overall export performance across the available years using export value and quantity.

Analytical Approach

- Examined the **year-wise distribution of export records** to assess data availability across the study period.
- Compared the number of records for each year from **2018 to 2024**.
- Identified variations in data coverage that should be considered during year-wise export performance analysis.

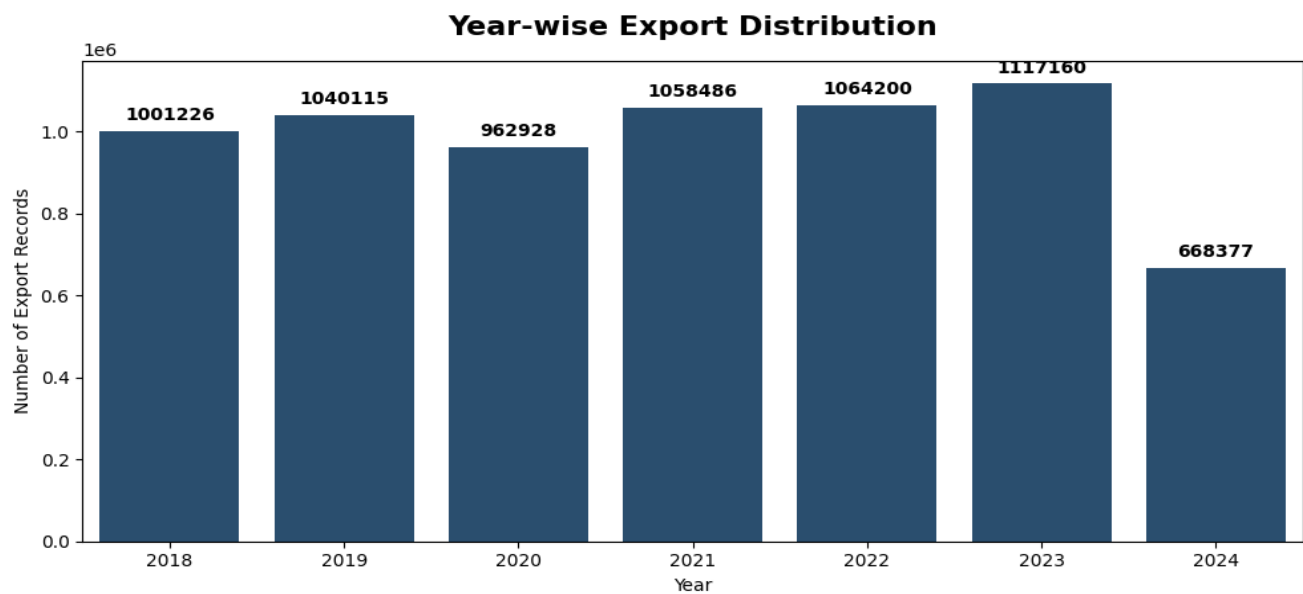


Figure 8.1: Year-Wise Distribution of Export Records (2018–2024)

Interpretation:

The chart shows the number of export records available for each year. Record coverage remains relatively consistent from **2018 to 2023**, while **2024 contains fewer records**.

Key Finding:

2023 recorded the highest number of export records (1,117,160), while 2024 recorded the lowest (668,377).

Business Insight:

The lower record count in **2024** should be considered when interpreting its export performance, as comparisons with previous years may be affected if the available data represents only a partial year.

8.2 Objective 2: Top-Performing Principal Commodities

Identify Top-Performing Commodities

Identify the principal commodities that contribute significantly to India's total export value.

Analytical Approach

- Analysed the **distribution of export records** across principal commodities to understand their representation in the dataset.
- Calculated the **total INR export value by commodity** to identify the principal commodities contributing most significantly to India's overall exports.
- Ranked commodities based on total export value and identified the **Top 10 commodities**.

Visualization 1 — Commodity Record Distribution

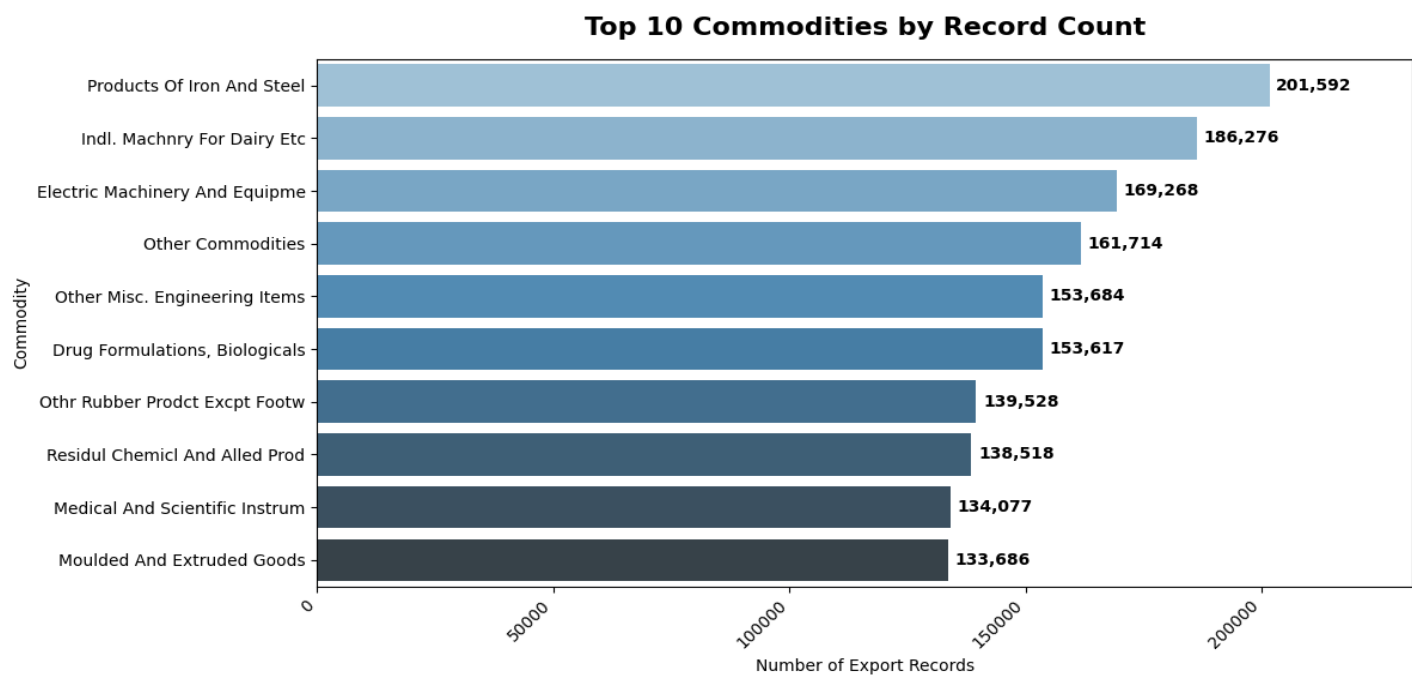


Figure 8.2: Distribution of Export Records by Principal Commodity

Interpretation:

The chart shows the frequency of export records across principal commodities. **Products Of Iron And Steel** has the highest representation with **201,592 records**, followed by **Indl. Machnry For Dairy Etc (186,276)** and **Electric Machinery And Equipme (169,268)**.

Key Finding:

Record frequency is highest for **Products Of Iron And Steel**, but record count alone does not indicate the commodity's total export-value contribution.

Visualization 2 — Top 10 Commodities by Export Value

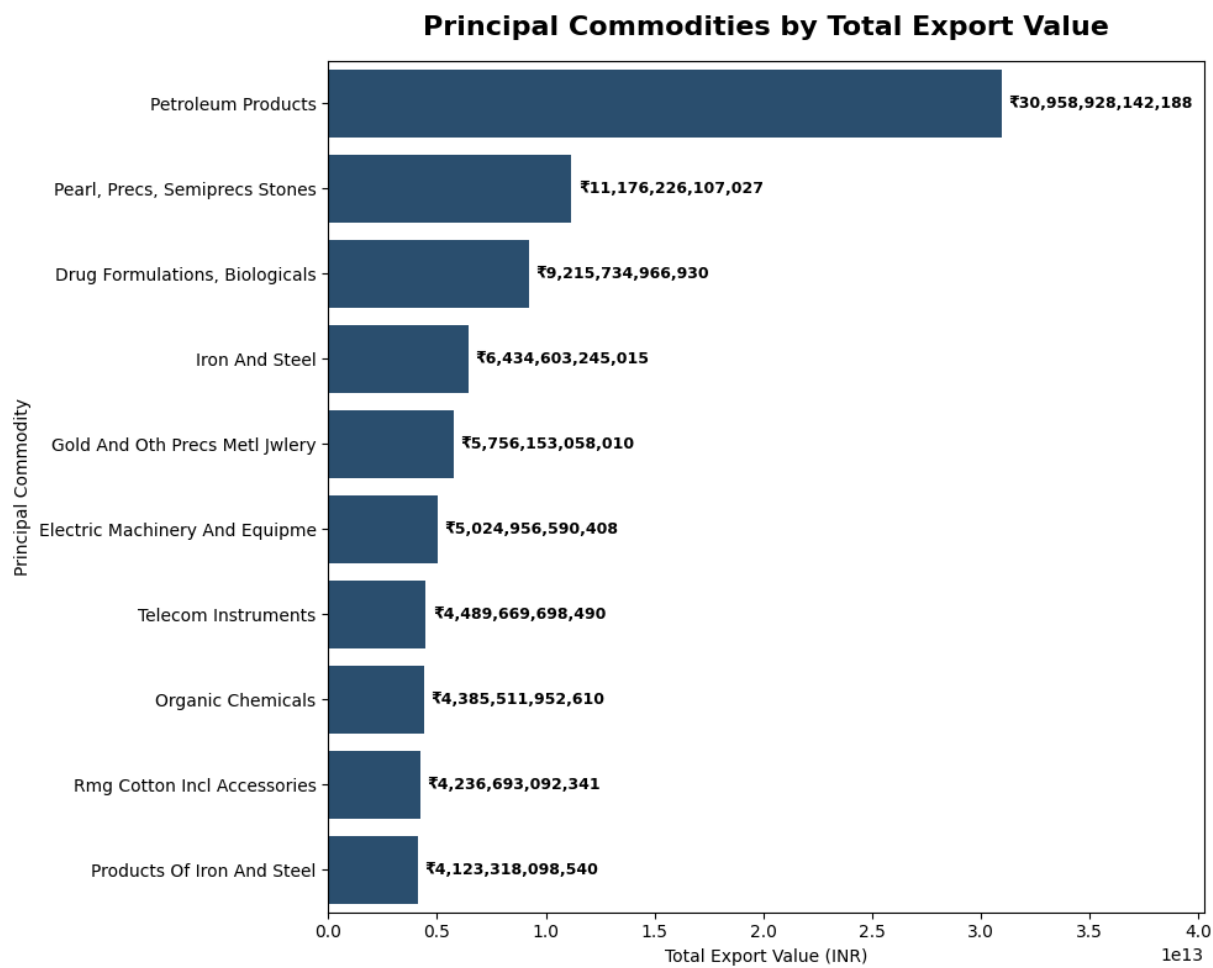


Figure 8.2: Top 10 Principal Commodities by Total Export Value (INR)

Interpretation:

The export-value analysis shows that **Petroleum Products** contributed the highest total export value at **₹30.96 trillion**, followed by **Pearl, Precs, Semiprecs Stones** (₹11.18 trillion) and **Drug Formulations, Biologicals** (₹9.22 trillion).

Key Finding:

Petroleum Products is the leading commodity by export value, with a substantially higher contribution than the other top-performing commodities.

Business Insight:

The comparison demonstrates that **commodity record frequency and economic contribution are not the same**. While *Products Of Iron And Steel* has the highest number of records, *Petroleum Products* generates the highest export value. Therefore, **export value is the more appropriate measure for identifying the major contributors to India's overall export performance**.

8.3 Objective 3: State-Wise Export Performance

Evaluate State-Wise Export Performance

Analyze export performance across states and identify the major exporting states.

Analytical Approach

- Analysed the number of export records across states to understand the distribution of export activity.
- Ranked states based on their number of recorded export transactions.
- Identified the major states with the highest representation in the dataset.

Visualization 1: State-Wise Distribution of Export Records

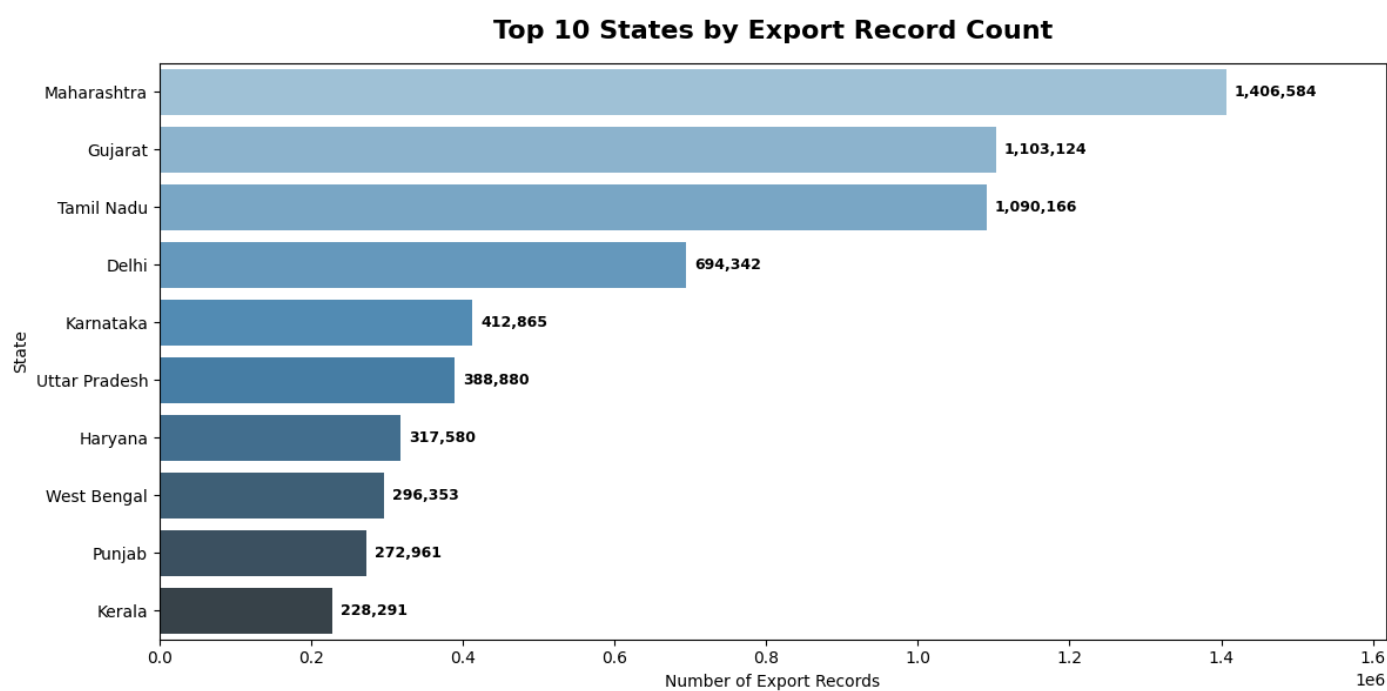


Figure 8.3: Distribution of Export Records by State-Wise

Interpretation:

The chart shows the distribution of export records across the top 10 states. **Maharashtra** has the highest number of records, followed by **Gujarat** and **Tamil Nadu**. The remaining states have comparatively fewer export records.

Key Finding:

Maharashtra recorded the highest number of export records (1,406,584), followed by Gujarat (1,103,124) and Tamil Nadu (1,090,166).

Business Insight:

The high record concentration in Maharashtra, Gujarat, and Tamil Nadu indicates that these states have a strong representation in the export dataset and are important states for further state-level export analysis.

Analytical Approach

- The total **export value in USD** was aggregated for each state across the 2018–2024 study period. States were then ranked based on their total export value, and the **Top 10 exporting states** were identified for comparison.
- **USD was selected as the primary measure because the analysis focuses on international trade. It provides a common reference currency for comparing export values across Indian states.**

Visualization 2 -Top 10 States by Total Export Value (USD), 2018–2024

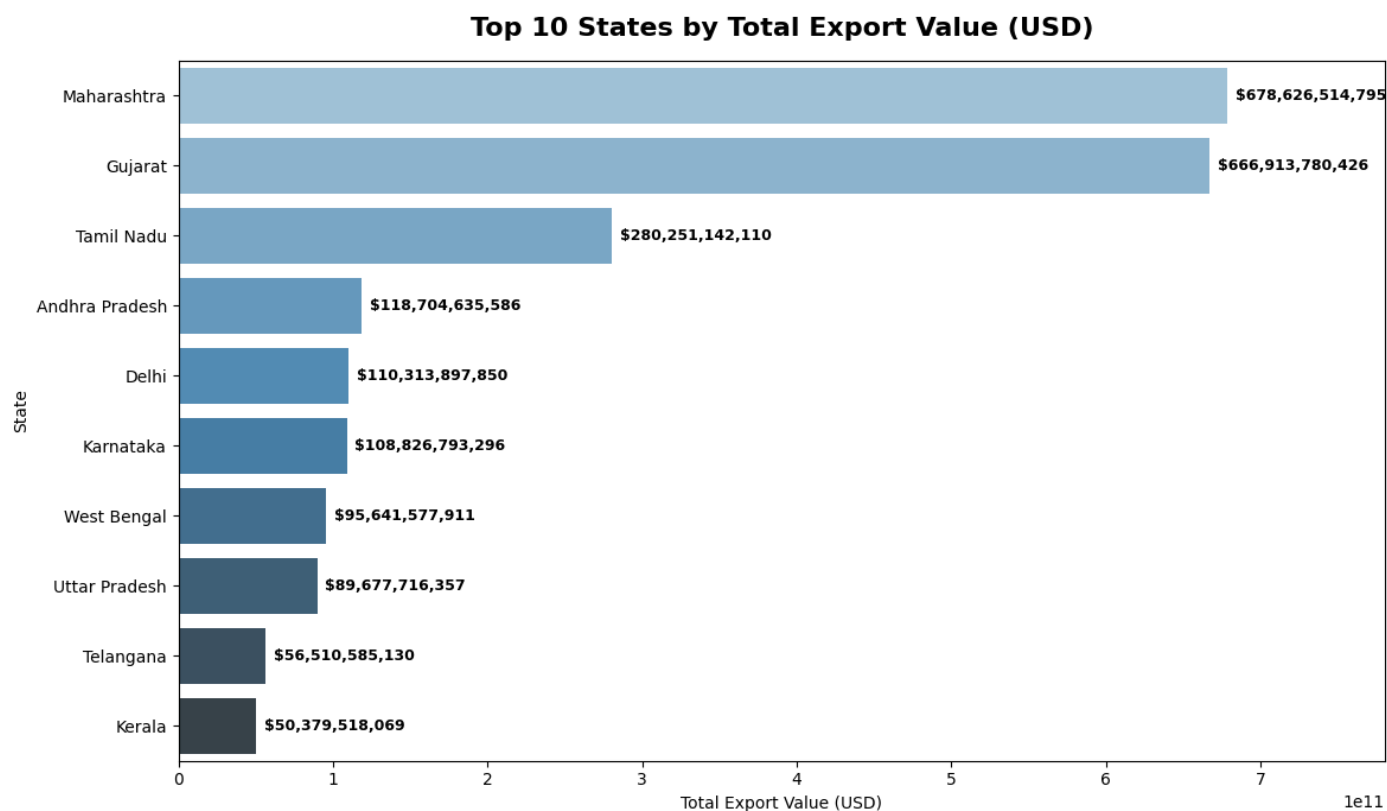


Figure 8.3: Top 10 States by Total Export Value (USD), 2018–2024

Interpretation:

The chart compares the total export value of the Top 10 states. **Maharashtra and Gujarat** recorded substantially higher export values than the other states, while **Tamil Nadu** ranked third. The remaining states contributed comparatively lower export values during the study period.

Key Finding:

Maharashtra recorded the highest total export value of **\$678.63 billion**, closely followed by **Gujarat** at **\$666.91 billion**. **Tamil Nadu** ranked third with **\$280.25 billion**. Andhra Pradesh, Delhi, and Karnataka followed with export values of **\$118.70 billion**, **\$110.31 billion**, and **\$108.83 billion**, respectively.

Business Insight:

The results indicate that export performance is **highly concentrated among a few major states**, with Maharashtra and Gujarat emerging as the leading contributors. This highlights the importance of these states in India's export ecosystem and provides a basis for further analysis of the commodities, ports, and markets driving their export performance.

8.4 Objective 4: Year-Wise Export Trends

Examine Yearly Export Trends

Track changes in export performance across years and identify periods of growth, decline, and significant variation.

Analytical Approach

The total **export value in USD** was aggregated for each year from **2018 to 2024**. The yearly values were compared to identify periods of growth, decline, and significant changes in India's export performance over the study period.

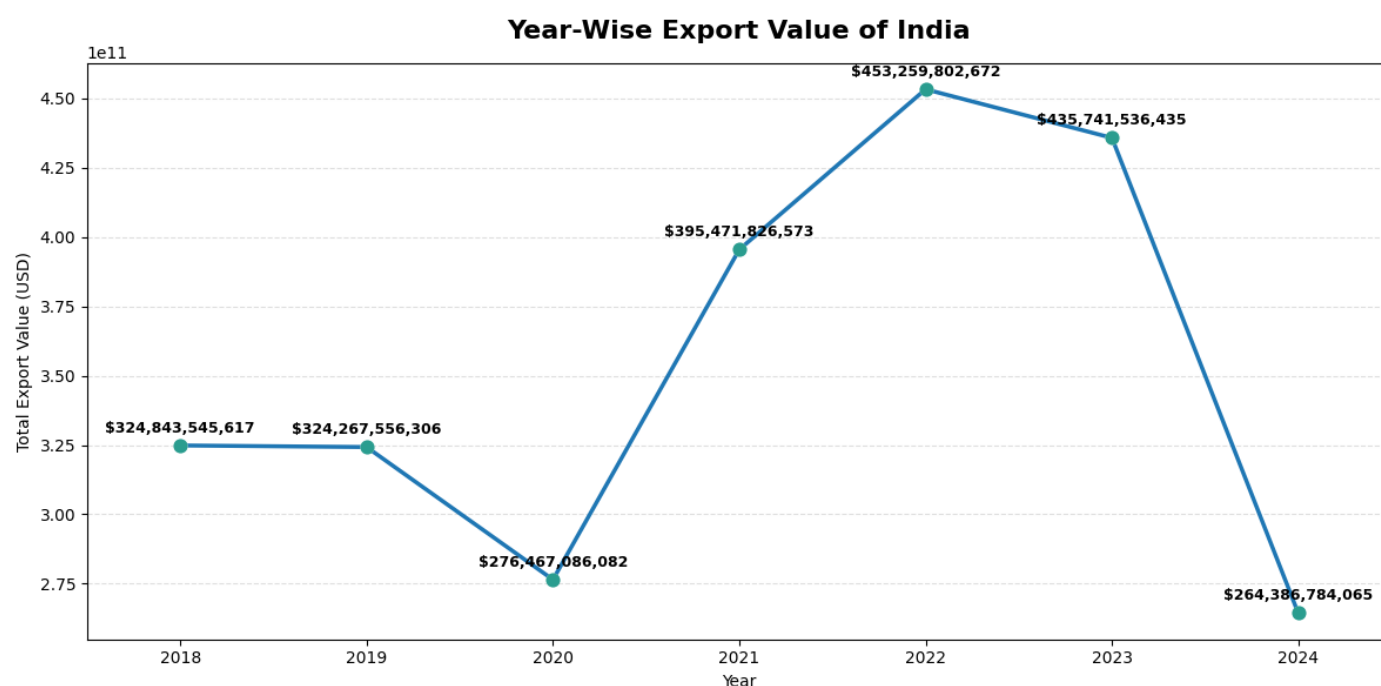


Figure 8.4: Year-Wise Export Value Trend (2018–2024)

Interpretation

The chart shows the movement of India's total export value across the study period. Export value remained relatively stable between **2018 and 2019**, declined in **2020**, and then increased significantly during **2021 and 2022**. Export value declined moderately in **2023** and recorded a substantial decrease in **2024**.

Key Finding

India's export value increased from **\$276.47 billion in 2020** to a peak of **\$453.26 billion in 2022**. It then declined to **\$435.74 billion in 2023** and further to **\$264.39 billion in 2024**, the lowest value in the study period.

Business Insight

The trend indicates significant variation in India's export performance across the study period. The strong recovery from **2020 to 2022** highlights a period of substantial export growth, while the decline after 2022 indicates a need to examine the commodities, states, and markets contributing to the change. The **2024 value should also be interpreted carefully if the dataset represents only a partial year**.

Data Consideration: The recorded export value for **2024 (\$264.39 billion)** was lower than the previous full-year values. However, the dataset contains only **7 months of data for 2024**. Therefore, the lower value should not be interpreted as an annual decline in India's export performance.

8.5 Objective 5: Commodity and State Relationship Analysis

Analyze Commodity–State Relationships

Examine the relationship between principal commodities and states to identify commodities that contribute significantly to exports from different states.

Analytical Approach

- Aggregated total **export value in USD** by **State** and **Principal Commodity**.
- Identified the **Top 20 commodities** based on overall export value.
- Created a **State × Commodity** matrix to compare the export contribution of major commodities across different states.

- Used a heatmap to identify states and commodities with relatively higher export values and observe concentration patterns.

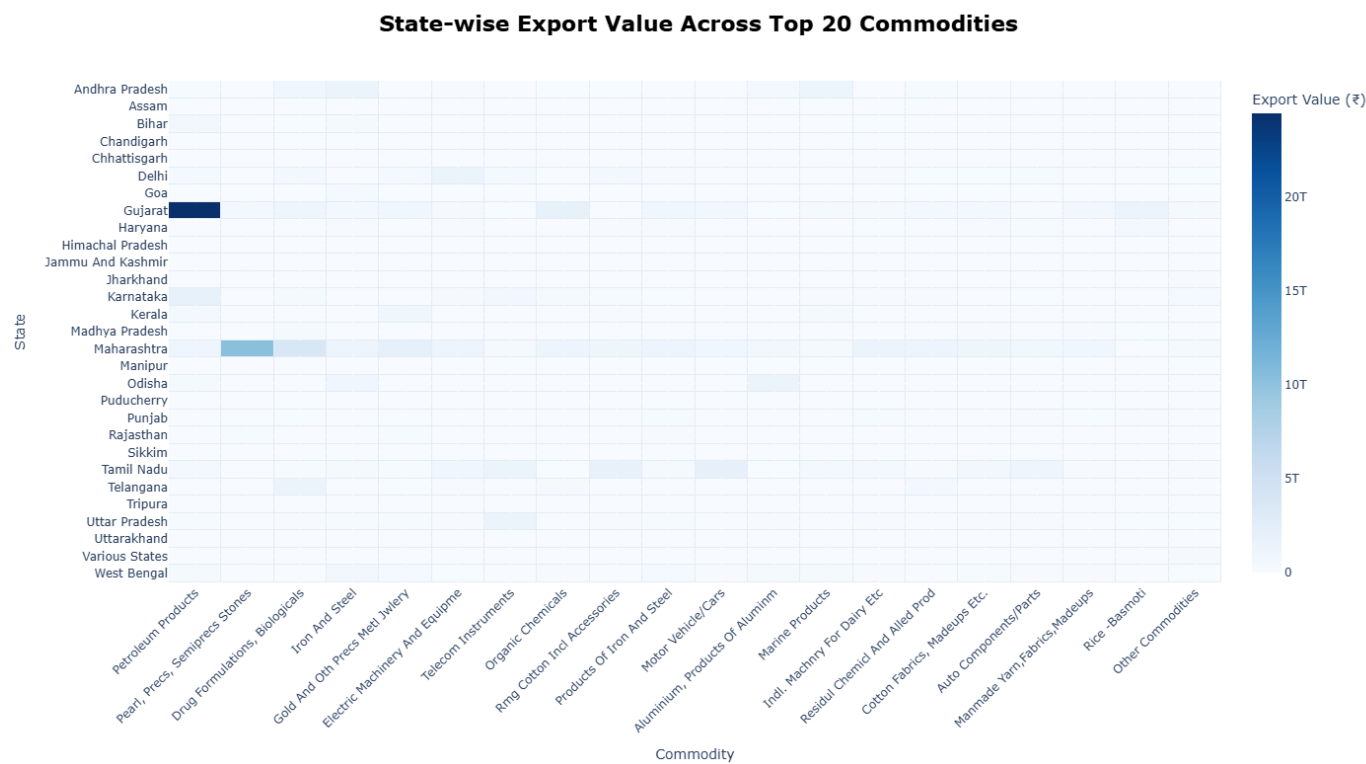


Figure 8.5: State-Wise Export Value Across Top 20 Principal Commodities

Interpretation

The heatmap presents the export value of the **Top 20 principal commodities across different states**. The variation in cell intensity highlights differences in commodity-level export contribution between states. It shows that certain commodities have relatively higher export values in specific states, indicating that state-level export performance is influenced by the commodity composition of each state.

Key Finding

The analysis shows that export activity is **not uniformly distributed across states and commodities**. Major commodities such as **Petroleum Products, Pearl and Precious Stones, Drug Formulations, Iron and Steel, and Gold and Precious Metal Jewellery** show significant export contributions across selected states, while other commodities have a more limited state-level presence.

Business Insight

The State–Commodity relationship highlights **commodity specialization across states**. Identifying which commodities contribute significantly to exports from particular states can help businesses and policymakers understand regional export strengths, prioritize relevant industries, and identify opportunities for commodity-specific export development.

8.6 Objective 6: Commodity-Wise Changes Across Years

Identify Changes in Commodity Performance

Analyze changes in commodity-wise export performance across different years.

Analytical Approach

- Analysed year-wise export value for major principal commodities to identify changes in their export performance over the study period.
- Compared commodity export values across **2018–2024** to identify increasing, decreasing, and fluctuating patterns.
- Focused on major commodities to make the year-wise comparison clear and meaningful.

Year-Wise Export Performance of Major Principal Commodities

A multi-series line chart can be used to compare the export value of selected major commodities across 2018–2024.

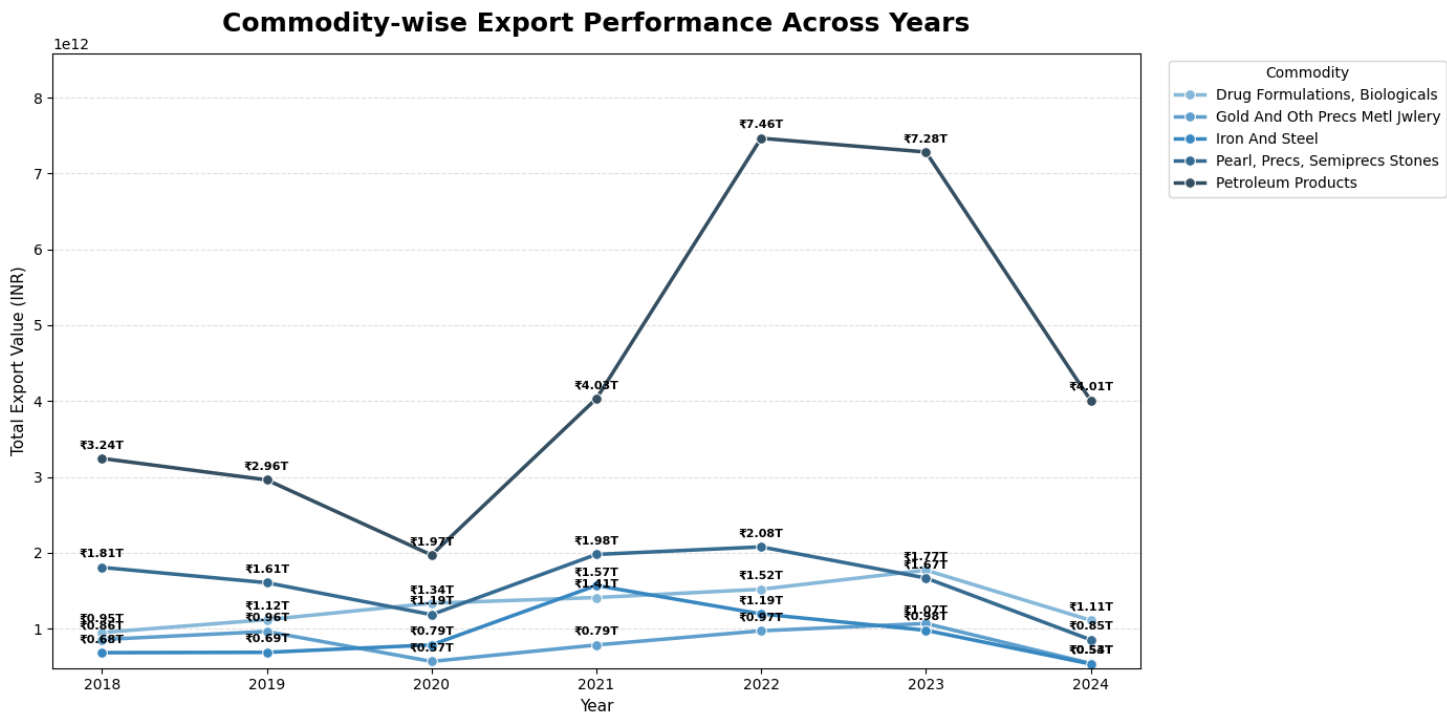


Figure 8.6: Year-Wise Export Performance of Selected Principal Commodities (2018–2024)

Interpretation

The visualization shows how the export performance of major principal commodities changed across the study period. Different commodities exhibit different patterns, with some showing relatively consistent performance while others experience noticeable increases, decreases, or fluctuations across years.

Key Finding

The analysis indicates that commodity export performance varies significantly across years, with major commodities contributing differently to India's export performance during the study period. This highlights that the composition of India's exports changes over time rather than remaining constant.

Business Insight

Year-wise commodity analysis helps identify commodities with sustained export performance, periods of strong growth, or declining contribution. These trends can support businesses and policymakers in identifying high-performing commodity segments and areas requiring further attention.

8.7 Objective 7: Contribution and Concentration Analysis

Measure Export Contribution and Concentration

Measure the contribution and concentration of major commodities and states within India's overall exports.

Analytical Approach

- Aggregated the total **export value in USD (dollars_value)** for each principal commodity.
- Calculated the **percentage contribution** of each commodity to India's total export value and ranked the commodities based on their contribution.
- Calculated the **cumulative contribution of the Top 20 commodities** to assess the concentration of export value among major commodities.
- State-wise contribution can be analysed using the same approach to identify the major states contributing to India's overall exports.

Visualization 1: Top 10 Commodities by Contribution to Total Export Value

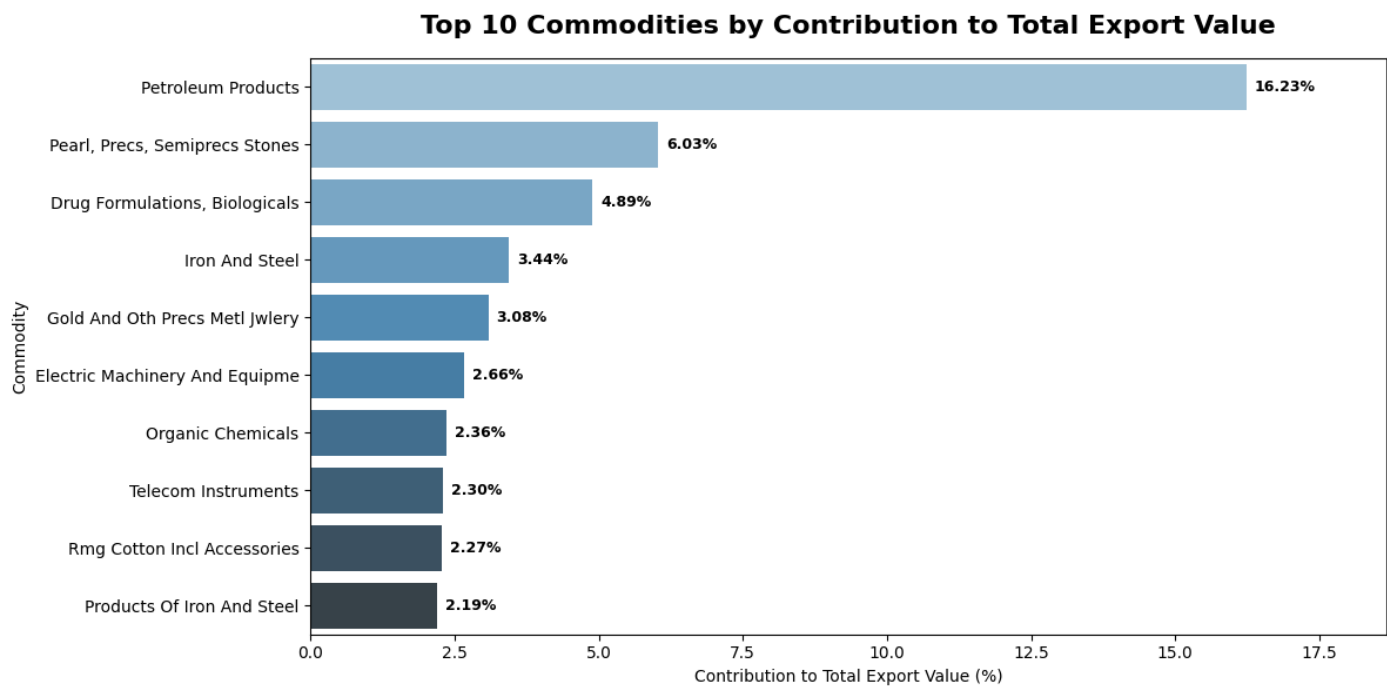


Figure 8.7 Top 10 Commodities by Contribution to Total Export Value

Visualization 1: Cumulative Contribution of Top 20 Commodities

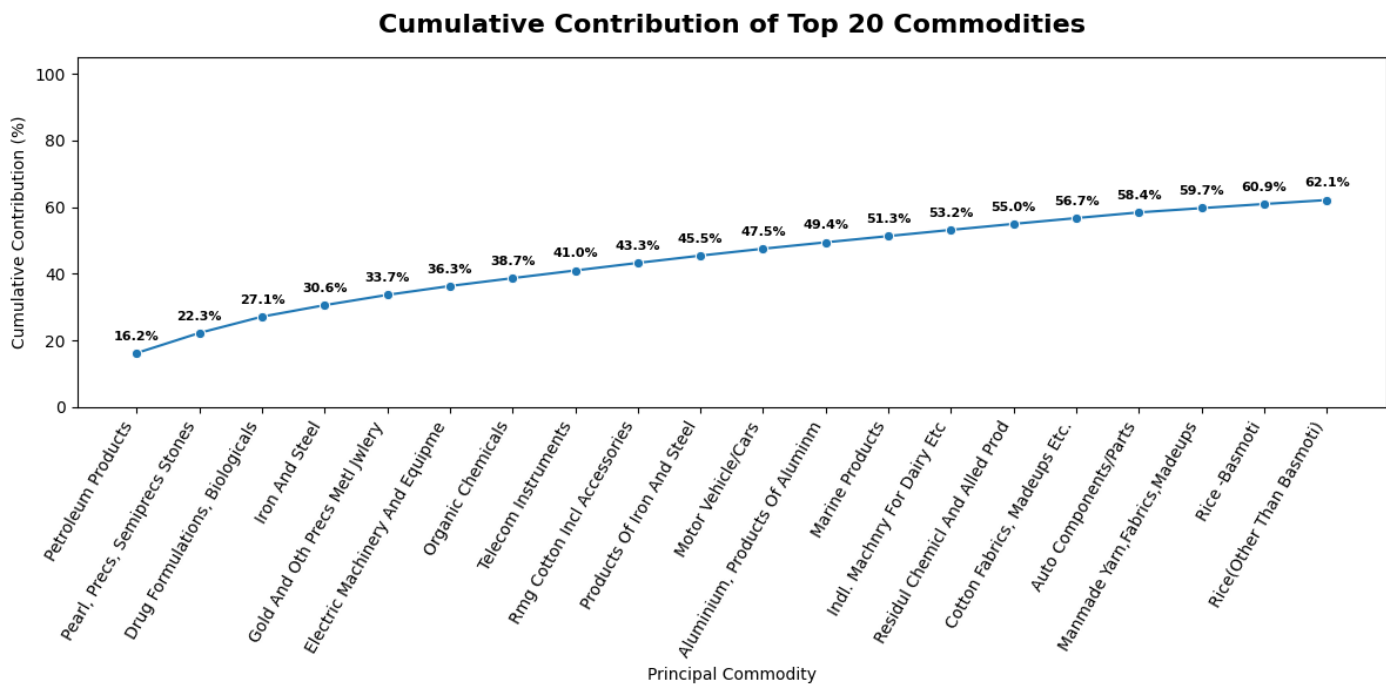


Figure 8.7 Cumulative Contribution of Top 20 Commodities

Interpretation

The analysis shows that India's export value is distributed unevenly across principal commodities. **Petroleum Products** has the highest individual contribution at **16.23%**, followed by **Pearl, Precs, Semiprecs Stones (6.03%)** and **Drug Formulations, Biologicals (4.89%)**.

The cumulative analysis shows that the **Top 20 commodities collectively contribute 62.12%** of India's total export value. This demonstrates that a relatively limited number of principal commodities account for a substantial share of India's overall exports.

Key Finding

Petroleum Products is the largest individual contributor, accounting for **16.23%** of total export value. The **Top 20 commodities together contribute 62.12%**, indicating a significant concentration of India's export value among major commodity categories.

Business Insight

The concentration of export value among a limited number of major commodities means that changes in the performance of these commodities can have a considerable impact on India's overall exports. Monitoring major contributors can help identify key export drivers and potential **diversification opportunities** to reduce dependence on a limited set of commodity categories.

9. Key Findings and Business Insights

- **Petroleum Products** was the largest export contributor, accounting for **16.23%** of total export value.
- The **Top 20 commodities contributed 62.12%** of total export value, indicating significant concentration among major commodities.
- **Maharashtra and Gujarat** were the leading states by total export value, followed by **Tamil Nadu**.
- Export value increased significantly from **2020 to 2022**, reaching the highest recorded value of **\$453.26 billion in 2022**.

- Major commodities showed considerable year-to-year variation, with **Petroleum Products** consistently remaining a key export contributor.
- The State–Commodity analysis revealed **different commodity strengths across states**, indicating regional specialization in exports.
- The findings highlight the importance of monitoring major export contributors while exploring **commodity and market diversification opportunities**.

10. Conclusion

The analysis provides a data-driven overview of India's export performance from **2018 to 2024** across commodities, states, and years. The findings show that export value is significantly influenced by a limited number of major commodities and states, with **Petroleum Products** being the leading commodity contributor and **Maharashtra and Gujarat** among the leading exporting states. Year-wise analysis also identified substantial changes in export performance, particularly between **2020 and 2022**. Overall, the project demonstrates how Python-based data analytics can transform large-scale export data into meaningful patterns and business insights.

11. Recommendations

- **Promote export diversification** to reduce dependence on a limited number of major commodities.
- **Strengthen high-performing commodities** by supporting sectors with strong and consistent export potential.
- **Leverage state-level strengths** by encouraging region-specific export development and infrastructure.
- **Monitor yearly commodity trends** to identify emerging opportunities and declining export categories.
- **Expand into new markets and products** to improve the resilience and long-term growth of India's exports.

12. Limitations of the Analysis

- The analysis is based on the available **2018–2024 dataset** and therefore reflects only the specified study period.
- **2024 contains only seven months of data**, so its values should not be compared directly with complete-year figures.
- The dataset contains a large number of records with **zero quantity**, which limits quantity-based interpretation for some analyses.
- The analysis is based on the variables available in the dataset and does not include external factors such as **global demand, exchange rates, trade policies, or economic conditions**.

13. Future Scope

- Extend the analysis using **complete and updated export data** for subsequent years.
- Incorporate **destination-country analysis** to identify major export markets and market opportunities.
- Perform **port-wise analysis** to evaluate major export hubs and infrastructure performance.
- Apply **predictive analytics and forecasting** to estimate future export trends.
- Develop an interactive **Power BI dashboard** to support continuous monitoring and business decision-making.